

Creating spatial filters

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```
library(CommuteCA)
library(dplyr) # A Grammar of Data Manipulation
library(fitdistrplus) # Help to Fit of a Parametric Distribution to Non-Censored or Censored Data
library(scales) # Scale data column-wise in a computationally efficient way
library(here) # enable easy file referencing in project-oriented workflows
library(ggplot2) # Create Elegant Data Visualizations Using the Grammar of Graphics
library(RColorBrewer) # color schemes for maps (and other graphics) designed
library(sf) # support for simple features, a standardized way to encode spatial vector data
library(tmap) # thematic maps
library(tidyr) # tidying data
library(matlib) # To get the eigenvectors
library(Hmisc) # To considerate weights when calculate the median
library(sf) # support for simple features, a standardized way to encode spatial vector data
library(spdep) # Construct neighbours list from polygon list
library(readr)
```

This file is an adaptation of an Rmarkdown from the **CommuteCA** package. CommuteCA was created in conjunction with the office of the *Research Data Center at McMaster University*, the *Sherman Centre for Digital Scholarship* and the *Mobilizing Justice*¹.

The **CommuteCA** R package was created to develop standardized methods for transport analysis in research, particularly for analysis using the *2021 Census of Population* from Statistics Canada. We focused our efforts on the *Commuting Reference Guide*, which provides valuable variables and information on commuting for the Canadian population aged 15 and older living in private households.

¹ For this demonstration, we will use the case of the city of Toronto. If you want to use the other options, the original data from the Census in an RDC office or the test data for all locations in Canada, *update* the address in the chunk.

Section objectives

After calculating accessibility, we will now advance our analysis by developing regression models to explore the relationship between job accessibility and individual outcomes. These models examine the connection between a dependent variable (an outcome) and predictor variables (job accessibility indicators and other socioeconomic information). They estimate the probability of the dependent variable, conditional on a linear combination of the predictor variables.

In social science, it is well understood that “correlation does not imply causality” [Cunningham, 2021]. Therefore, it is important to evaluate the actual impact of a predictor variable (the cause) on the outcome (the effect). This requires studying the process that deter-

mines the cause-and-effect relationship between variables, a process known as *causal inference*.

Including causal inference in our modeling approach means we must account for the possible (and indeed likely) presence of *endogeneity*. Endogeneity occurs when the effect of an independent variable on a dependent variable cannot be interpreted causally, often due to omitted variables that lead to biased (i.e., inconsistent) estimates [Antonakis et al., 2010]. There are various ways to address endogeneity. In our case, we propose using an instrumental variable.

In this section, we will create synthetic instrumental variables obtained after applying a spatial filtering to the spatial availability variables created in the previous sections, in a methodology created by Gallo and Páez [2013].

Suggested Readings

- Le Gallo, J., & Páez, A. (2013). Using synthetic variables in instrumental variable estimation of spatial series models. *Environment and Planning A*, 45(9), 2227-2242.
- Cunningham, S. (2021). *Causal inference: The mixtape*. Yale university press.

Instrumental variables

A fundamental assumption in regression analysis is the absence of correlation between the explanatory variables and the error terms [Le Gallo and Páez, 2013]. Endogeneity occurs when an independent variable correlates with the regression error term. To predict outcomes from the 2021 Census of Population, we propose to address endogeneity by applying an instrumental variable technique. When an endogenous variable is included in a regression model, the consistency of the estimators is compromised, affecting the results and hindering the ability to establish a cause-effect relationship between the explanatory and dependent variables.

An instrumental variable Z must satisfy the following criteria [Cunningham, 2021, Luz and Portugal, 2022]:

1. Z must be highly correlated with and have a causal effect on the endogenous variable D, or share a common cause with D.
2. Z must affect the dependent variable Y only through D, with no direct effect of Z on Y.
3. Z must not be correlated with the regression residuals of D on Y.

A good instrument for the proposed models must be highly correlated with and have a causal effect (or share a common cause) on the

accessibility variable (the endogenous variable); affect employment outcomes only through accessibility; and not share common causes with employment outcomes. In simpler terms, the instrument must be correlated with employment outcomes only through accessibility.

Examples of instrumental variables used in studies analyzing the relationship between employment outcomes and job accessibility include Euclidean distance from the spatial unit centroid to the nearest major road or employment subcenter [Delmelle et al., 2021, Jin and Paulsen, 2018], share of no-vehicle households [Merlin and Hu, 2017], transit accessibility and car unavailability [Johnson et al., 2017], distance to rivers [Duarte et al., 2023], and population density ([Bastiaanssen et al., 2021].

However, two important challenges arise in the search for relevant instruments:

1. The likelihood that the instruments and error terms are correlated increases as the correlation between the endogenous variable and the instruments becomes stronger, which can render the instruments invalid.
2. Instruments that are only weakly correlated with the endogenous variable may be ineffective and perform poorly.

In response to these challenges, Le Gallo and Páez [2013] developed an alternative, cutting-edge approach that relies on synthetic instrumental variables when the model involves spatial data. The researchers demonstrated that synthetic variables can be generated using spatial filtering, a technique that removes spatial residual autocorrelation to improve regression analysis. They propose the use of an eigenvector-based technique, in which transforms a projection of the contiguity matrix into latent map patterns through eigenvector structural analysis. These latent patterns, included in the spatial structure of the system, are then used to generate synthetic variables, which serve as instruments in IV estimation. Since they are derived from the spatial structure of the system, these synthetic instruments are exogenous, thereby fulfilling the core requirement of IV estimation.

We have chosen to use this technique because, given our aim to create a methodology that can be applied to any Canadian city, there are no universally applicable candidates for instrumentation that could be obtained in every context.

Spatial Filtering

According to Le Gallo and Páez [2013], spatial filtering is a method to deal with, or filter from the residuals, spatial autocorrelation present

during regression analysis. Spatial autocorrelation is the degree of similarity among neighboring observation. Usually, the definition of the neighboring (ie, spatial structure) is made by the use of spatial weight matrix W , with individual values represented by:

$$w_{ij} = \begin{cases} w_{ij} > 0, & \text{if } i \text{ and } j \text{ are neighbors,} \\ w_{ij} = 0, & \text{otherwise.} \end{cases}$$

Neighborhood can be defined on the basis of contiguity, distance, or length of shared edge, among other criteria. So, having a weighted matrix that represents the spatial structure, the steps to generate a filter can be defined as follows:

1. Build a spatial weights matrix to capture the distances between all pairs of spatial units (in our case, Dissemination Areas, or DAs) under analysis.
2. Select a distance threshold to identify neighbors for each DA. This process usually results in a binary spatial contiguity matrix. There is no clear theoretical guidance on the optimal distance threshold for constructing the binary contiguity matrix.
3. Compute all eigenvectors associated with the contiguity matrix. These eigenvectors are orthogonal to one another and each represents a portion of the variance in the spatial contiguity matrix. Each independent variable will have its own set of associated eigenvectors.
4. Construct the spatial filter as a linear combination of a subset of these eigenvectors. The spatial filter for each independent variable is typically obtained by regressing the independent variable on the subset of eigenvectors with p-values below a specified threshold (for instance, $p = 0.05$), and then using the predicted values of the independent variable as the synthetic instrument.

The pseudo-code to build the spatial filter is as follows:

- Initialize an index value $i = 1$, an empty vector for the spatial filter $S = 0$, and a constant vector $X = 1$.
- In a for-loop and for every $i \leq n$, with n being number of eigenvectors, select the eigenvector E_i obtained from the weighted matrix as a candidate for inclusion in the spatial filter, and estimate the following model using Ordinary Least Squares (OLS), where θ and β are vectors of coefficients, and ϵ is a vector of error terms.

$$Y = \theta X + \beta E_i + \epsilon$$

- If coefficient i is significant at a pre-determined level (eg, $p\text{-value} \leq 0.05$), then synthesize the eigenvector and the existing filter: $S = S + \beta E_i$.

- The spatial filter S is the synthetic instrumental variable.

In summary, a spatial filter will be created to generate a synthetic counterpart that accurately reproduces a spatial random variable.

Let's code!

Load the packages:

Read and preprocess the files

```
code <- c("3518", # Durham
         "3519", # York
         "3520", # Toronto
         "3521", # Peel
         "3524", # Halton
         "3525" # Hamilton
        )
```

```
code_label <- "GTAH"
```

- Accessibility measures
- Spatial Availability :

Folder with the accessibility files

```
measures_folder <- paste0(here::here(), "/data-raw/output/", code_label, "/CT/accessibility-measures/")
```

Read file job spatial availability by mode

```
SA_mode <- read.csv(paste0(measures_folder, "SA_mode.csv"),
                   colClasses = c("character", "factor", "numeric", "numeric", "numeric")) %>%
  dplyr::select(-c(SA_nonc_im, labour_force)) %>%
  pivot_wider(names_from = PwMode_label, values_from = SA_im, names_sep = "_", values_fill = 0) %>%
  mutate(Active = Walk + Bike) %>%
  rename(CTUID = from_id)
```

```
SA_general <- read.csv(paste0(measures_folder, "SA_general.csv"),
                     colClasses = c("character", "numeric")) %>%
  rename(CTUID = from_id, General = SA_i) %>%
  dplyr::select(-c(SA_nonc_i))
```

```
SA <- SA_mode %>%
  left_join(SA_general, by = "CTUID") %>%
  pivot_longer(cols = c(Bike, Walk, `Car-motor`, Transit, Active, General),
              names_to = "PwMode_label",
              values_to = "SA_im")
```

Preparing the accessibility file:

```
grid_expanded <- expand.grid(CTUID = unique(SA$CTUID),
                             PwMode_label = unique(SA$PwMode_label))

SA_final <- grid_expanded %>%
  left_join(SA, by = c("CTUID", "PwMode_label")) %>%
  mutate(SA_im = replace_na(SA_im, 0)) %>%
  dplyr::distinct(CTUID, PwMode_label, SA_im)
```

- Spatial Availability per person:

```
# Read file job spatial availability by mode
SA_pp_mode <- SA_pp <- read_csv(paste0(measures_folder, "SA_pp_long.csv"),
                                col_types = cols(
                                  from_id = col_character()))

SA_pp <- SA_pp_mode %>%
  rename(CTUID = from_id, SA_pp_im = SA_pp)
```

Preparing the accessibility file:

```
SA_final_pp <- grid_expanded %>%
  left_join(SA_pp, by = c("CTUID", "PwMode_label")) %>%
  mutate(SA_pp_im = replace_na(SA_pp_im, 0)) %>%
  dplyr::distinct(CTUID, PwMode_label, SA_pp_im)
```

- Hansen Type:

```
# Read file HT by mode
HT_mode <- read_csv(paste0(measures_folder, "HT_mode.csv"),
                    colClasses = c("character", "factor", "numeric")) %>%
  pivot_wider(names_from = PwMode_label, values_from = HT_im, names_sep = "_", values_fill = 0) %>%
  mutate(Active = Walk + Bike) %>%
  rename(CTUID = from_id)

HT_general <- read_csv(paste0(measures_folder, "HT_general.csv"),
                       colClasses = c("character", "numeric")) %>%
  rename(CTUID = from_id, General = HT_i)

HT <- HT_mode %>%
  left_join(HT_general, by = "CTUID") %>%
  pivot_longer(cols = c(Bike, Walk, `Car-motor`, Transit, Active, General),
               names_to = "PwMode_label",
               values_to = "HT_im")
```

Preparing the accessibility file:

```
grid_expanded <- expand.grid(CTUID = unique(HT$CTUID),
                             PwMode_label = unique(HT$PwMode_label))
```

```
HT_final <- grid_expanded %>%
  left_join(HT, by = c("CTUID", "PwMode_label")) %>%
  mutate(HT_im = replace_na(HT_im, 0))
```

- Spatial files

Reading the dissemination areas spatial file:

```
# Folder
directory_spatial_files <- paste0(here(), "/data-raw/output/", code_label, "/spatial-files/") # Update it

# Read file
ct_file <- st_read(paste0(directory_spatial_files, "census_tracts.gpkg"))

## Reading layer `census_tracts' from data source
##   `C:\Bruno\NSTP\data-raw\output\GTAH\spatial-files\census_tracts.gpkg'
##   using driver `GPKG'
## Simple feature collection with 1489 features and 11 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -80.24849 ymin: 43.05052 xmax: -78.43698 ymax: 44.39486
## Geodetic CRS:   WGS 84
```

Reading the census_divisions geometries:

```
census_subdivisions <- st_read(paste0(directory_spatial_files, "census_subdivisions.gpkg"))

## Reading layer `census_subdivisions' from data source
##   `C:\Bruno\NSTP\data-raw\output\GTAH\spatial-files\census_subdivisions.gpkg'
##   using driver `GPKG'
## Simple feature collection with 25 features and 7 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: -80.24849 ymin: 43.05054 xmax: -78.43698 ymax: 44.43323
## Geodetic CRS:   WGS 84

census_subdivisions <- ngeo::st_remove_holes(census_subdivisions)
```

Creating folder to save files

Creating a directory to export the figures and tables:

```
diretorio_export_figures <- paste0(here(), "/data-raw/output/", code_label, "/CT/spatial-filter-figures/")

if(!dir.exists(diretorio_export_figures)){
```

```

  dir.create(diretorio_export_figures, recursive = TRUE)}

diretorio_export_tables <- paste0(here(), "/data-raw/output/", code_label, "/CT/spatial-filter-measures/")

if(!dir.exists(diretorio_export_tables)){
  dir.create(diretorio_export_tables, recursive = TRUE)}

```

Spatial filters

The first spatial filter we will create is based on contiguity. This means that the weighted matrix will only have values of 0 or 1, where 0 indicates that the DAs are not neighbors and 1 indicates that they are neighbors.

Spatial filter based on the contiguity

First, we will create the contiguity matrix:

```

# Contiguity matrix
mtx_distance_binary <- spdep::nb2mat(spdep::poly2nb(ct_file), style = "B", zero.policy = TRUE)

# Changing the diagonal column to 1
diag(mtx_distance_binary) <- 1

# Renaming the columns and row index so we can identify each DA
rownames(mtx_distance_binary) <- ct_file$CTUID
colnames(mtx_distance_binary) <- ct_file$CTUID

```

Calculating the contiguity matrix eigenvectors:

```

ev_dist <- eigen(mtx_distance_binary)

# Creating an Eigenvectors Data Frame
eigv_df <- as.data.frame(Re(ev_dist$vectors))

colnames(eigv_df) <- paste0("EV", seq_len(ncol(eigv_df)))

# Adding the DA code to the Eigenvectors table
weight_matrix_eigv_dist <- cbind(eigv_df, CTUID = as.character(rownames(mtx_distance_binary)))

```

Spatial Availability

```

modes <- unique(SA_final$PwMode_label)

mode_results_dist <- list()

```



```

for (mode in modes) {
  cat("Creating the spatial filter (distance) for the transportation mode:", mode, "\n")

  # Creating a df with the accessibility values
  y <- SA_final %>%
    filter(PwMode_label == mode) %>%
    left_join(weight_matrix_eigv_dist, by = c("CTUID" = "CTUID")) %>%
    mutate(x = 1, Sf_dist_SA = 0)

  # Creating the spatial filter (Sf_dist_SA)
  for (ev_name in names(y)[startsWith(names(y), "EV")]) {
    formula <- as.formula(paste("SA_im ~ x + Sf_dist_SA +", ev_name))
    model <- lm(formula, data = y)
    coefs <- summary(model)$coefficients

    if (ev_name %in% rownames(coefs)) {
      p_value <- coefs[ev_name, "Pr(>|t|)"]
      B_value <- coefs[ev_name, "Estimate"]

      if (!is.na(p_value) && p_value < 0.05) {
        y$Sf_dist_SA <- y$Sf_dist_SA + B_value * y[[ev_name]]
      }
    }
  }

  mode_results_dist[[mode]] <- y
}

## Creating the spatial filter (distance) for the transportation mode: Bike
## Creating the spatial filter (distance) for the transportation mode: Walk
## Creating the spatial filter (distance) for the transportation mode: Car-motor
## Creating the spatial filter (distance) for the transportation mode: Transit
## Creating the spatial filter (distance) for the transportation mode: Active
## Creating the spatial filter (distance) for the transportation mode: General

Creating a data frame with all spatial filters and accessibility:

spatial_filter_dist <- as.data.frame(bind_rows(mode_results_dist, .id = "PwMode_label")) %>%
  dplyr::select(CTUID, PwMode_label, SA_im, Sf_dist_SA)

Pearson correlation between job spatial availability and spatial filter
by transportation mode:

correlations <- spatial_filter_dist %>%
  group_by(PwMode_label) %>%
  summarise(correlation = cor(SA_im, Sf_dist_SA, use = "complete.obs"))

```

correlations

```
## # A tibble: 6 x 2
##   PwMode_label correlation
##   <chr>           <dbl>
## 1 Active           0.909
## 2 Bike             0.931
## 3 Car-motor        0.794
## 4 General          0.788
## 5 Transit          0.873
## 6 Walk             0.888
```

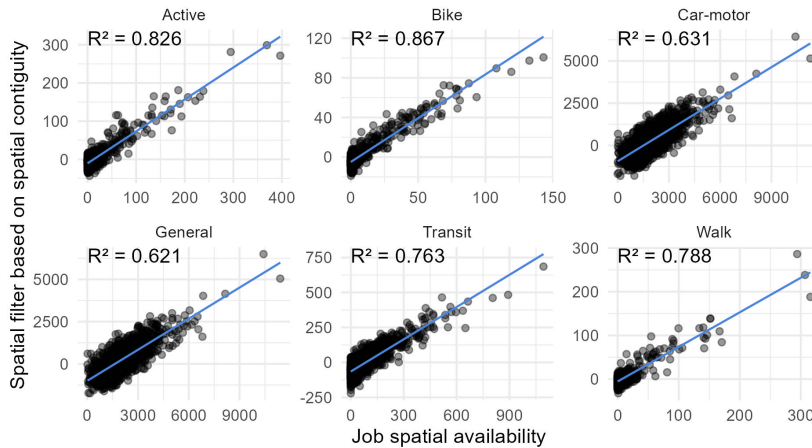
Creating a scatter plot figure:

```
r2_labels <- spatial_filter_dist %>%
  group_by(PwMode_label) %>%
  summarise(r2 = summary(lm(Sf_dist_SA ~ SA_im))$r.squared) %>%
  mutate(label = paste0("R2 = ", round(r2, 3)))

spatial_filter_dist_fig <- ggplot(spatial_filter_dist, aes(x = SA_im, y = Sf_dist_SA)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "#4b84db", linewidth = 0.55) +
  geom_text(data = r2_labels,
            aes(x = -Inf, y = Inf, label = label),
            hjust = -0.1, vjust = 1.1, inherit.aes = FALSE) +
  facet_wrap(~ PwMode_label, scales = "free") +
  theme_minimal() +
  labs(x = "Job spatial availability", y = "Spatial filter based on spatial contiguity") +
  theme_minimal()

# Saving figure
ggsave(file = paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA.jpg"),
        plot = spatial_filter_dist_fig,
        width = 16,
        height = 9,
        units = "cm",
        dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA.jpg"), )
```



Visualizing the spatial filter in form of maps:

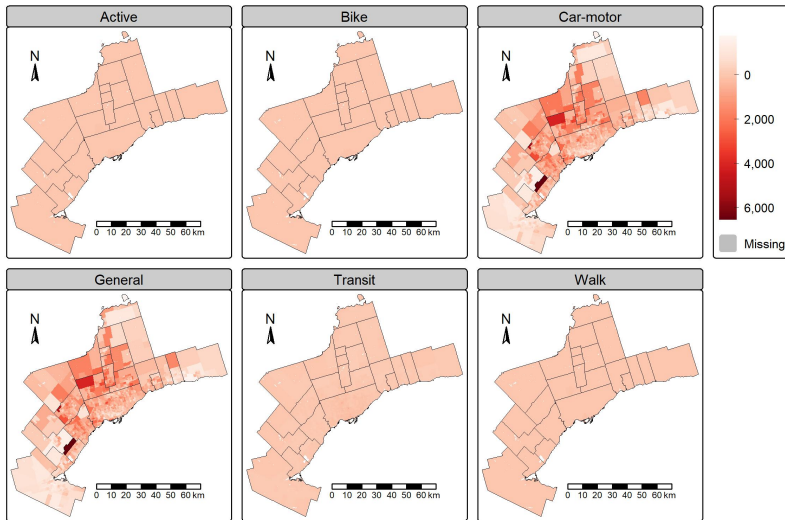
```
spatial_filter_contiguity_spatial <- grid_expanded %>%
  filter(!is.na(PwMode_label)) %>%
  left_join(spatial_filter_dist, by = c("CTUID" = "CTUID",
                                       "PwMode_label" = "PwMode_label")) %>%
  left_join(ct_file %>% mutate(CTUID = as.character(CTUID)), by = c("CTUID" = "CTUID"))

spatial_filter_contiguity_spatial <- st_as_sf(spatial_filter_contiguity_spatial)

spatial_filter_contiguity_fig <- spatial_filter_contiguity_spatial %>%
  tm_shape(spatial_filter_contiguity_spatial) +
  tm_polygons("Sf_dist_SA",
             style = "cont",
             palette = "Reds",
             title = " ",
             border.col = NULL) +
  tm_facets(by = "PwMode_label") +
  tm_scalebar(position = c("right", "bottom")) +
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(spatial_filter_contiguity_fig,
          paste0(diretorio_export_figures, "contiguity_spatial_filter_SA.jpg"),
          width = 20,
          height = 18,
          units = "cm",
          dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures, "/contiguity_spatial_filter_SA.jpg"))
```



Spatial Availability Per Person

```
mode_results_dist <- list()

for (mode in modes) {
  cat("Creating the spatial filter (distance) for the transportation mode:", mode, "\n")

  # Creating a df with the accessibility values
  y <- SA_final_pp %>%
    filter(PwMode_label == mode) %>%
    left_join(weight_matrix_eigv_dist, by = c("CTUID" = "CTUID")) %>%
    mutate(x = 1, Sf_dist_SA_pp = 0)

  # Creating the spatial filter (Sf_dist_SA_pp)
  for (ev_name in names(y)[startsWith(names(y), "EV")]) {
    formula <- as.formula(paste("SA_pp_im ~ x + Sf_dist_SA_pp +", ev_name))
    model <- lm(formula, data = y)
    coefs <- summary(model)$coefficients

    if (ev_name %in% rownames(coefs)) {
      p_value <- coefs[ev_name, "Pr(>|t|)"]
      B_value <- coefs[ev_name, "Estimate"]
    }
  }
}
```

```

    if (!is.na(p_value) && p_value < 0.05) {
      y$Sf_dist_SA_pp <- y$Sf_dist_SA_pp + B_value * y[[ev_name]]
    }
  }
}

mode_results_dist[[mode]] <- y
}

## Creating the spatial filter (distance) for the transportation mode: Bike
## Creating the spatial filter (distance) for the transportation mode: Walk
## Creating the spatial filter (distance) for the transportation mode: Car-motor
## Creating the spatial filter (distance) for the transportation mode: Transit
## Creating the spatial filter (distance) for the transportation mode: Active
## Creating the spatial filter (distance) for the transportation mode: General

Creating a data frame with all spatial filters and accessibility:

spatial_filter_dist_SA_pp <- as.data.frame(bind_rows(mode_results_dist, .id = "PwMode_label")) %>%
  dplyr::select(CTUID, PwMode_label, SA_pp_im, Sf_dist_SA_pp)

Pearson correlation between job spatial availability and spatial filter
by transportation mode:

correlations <- spatial_filter_dist_SA_pp %>%
  group_by(PwMode_label) %>%
  summarise(correlation = cor(SA_pp_im, Sf_dist_SA_pp, use = "complete.obs"))

correlations

## # A tibble: 6 x 2
##   PwMode_label correlation
##   <chr>          <dbl>
## 1 Active          0.880
## 2 Bike           0.962
## 3 Car-motor      0.953
## 4 General        0.939
## 5 Transit        0.927
## 6 Walk           0.982

Creating a scatter plot figure:

r2_labels <- spatial_filter_dist_SA_pp %>%
  group_by(PwMode_label) %>%
  summarise(r2 = summary(lm(Sf_dist_SA_pp ~ SA_pp_im))$r.squared) %>%
  mutate(label = paste0("R² = ", round(r2, 3)))

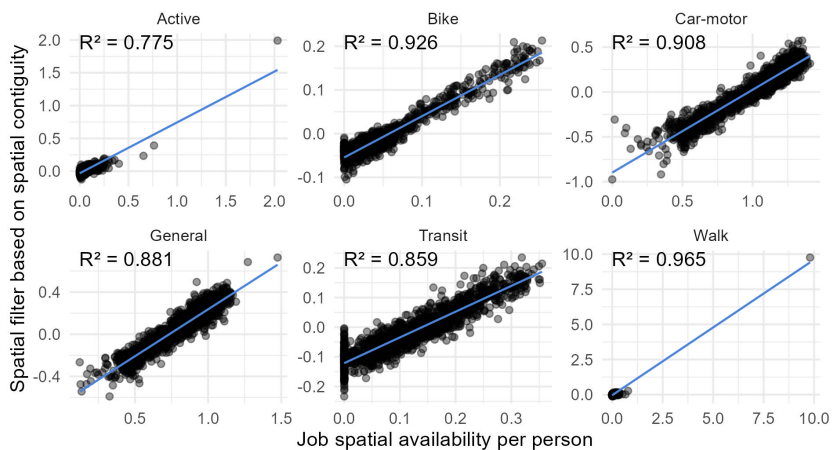
```

```
spatial_filter_dist_fig_SA_pp <- ggplot(spatial_filter_dist_SA_pp, aes(x = SA_pp_im, y = Sf_dist_SA_pp)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "#4b84db", linewidth = 0.55) +
  geom_text(data = r2_labels,
            aes(x = -Inf, y = Inf, label = label),
            hjust = -0.1, vjust = 1.1, inherit.aes = FALSE) +
  facet_wrap(~ PwMode_label, scales = "free") +
  theme_minimal() +
  labs(x = "Job spatial availability per person", y = "Spatial filter based on spatial contiguity") +
  theme_minimal()
```

Saving figure

```
ggsave(file = paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA_pp.jpg"),
       plot = spatial_filter_dist_fig_SA_pp,
       width = 16,
       height = 9,
       units = "cm",
       dpi = 300)
```

```
knitr::include_graphics(paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA_pp.jpg"))
```



Visualizing the spatial filter in form of maps:

```
spatial_filter_contiguity_spatial_SA_pp <- grid_expanded %>%
  filter(!is.na(PwMode_label)) %>%
  left_join(spatial_filter_dist_SA_pp, by = c("CTUID" = "CTUID",
                                             "PwMode_label" = "PwMode_label")) %>%
  left_join(ct_file %>% mutate(CTUID = as.character(CTUID)), by = c("CTUID" = "CTUID"))

spatial_filter_contiguity_spatial_SA_pp <- st_as_sf(spatial_filter_contiguity_spatial_SA_pp)

spatial_filter_contiguity_fig_SA_pp <- spatial_filter_contiguity_spatial_SA_pp %>%
  tm_shape(spatial_filter_contiguity_spatial_SA_pp) +
```

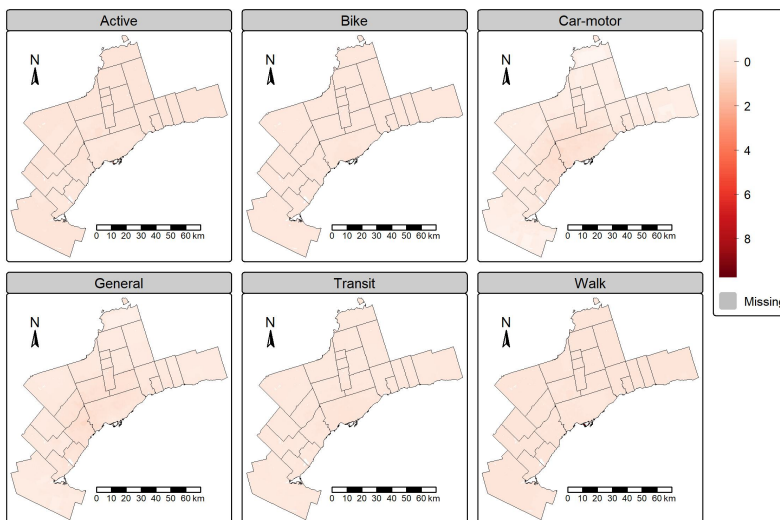
```

tm_polygons("Sf_dist_SA_pp",
  style = "cont",
  palette = "Reds",
  title = " ",
  border.col = NULL) +
tm_facets(by = "PwMode_label") +
tm_scalebar(position = c("right", "bottom")) +
tm_compass(position = c("left", "top"), size = 1.0) +
tm_shape(census_subdivisions) +
tm_borders("black", lwd=0.5)

tmap_save(spatial_filter_contiguity_fig_SA_pp,
  paste0(diretorio_export_figures,"contiguity_spatial_filter_SA_pp.jpg"),
  width = 20,
  height = 18,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"/contiguity_spatial_filter_SA_pp.jpg"))

```



Creating a scatter plot figure:

```

spatial_filter_dist_SA_pp_3m <- spatial_filter_dist_SA_pp %>%
  filter(PwMode_label %in% c("Active","Transit","Car-motor"))

```

```

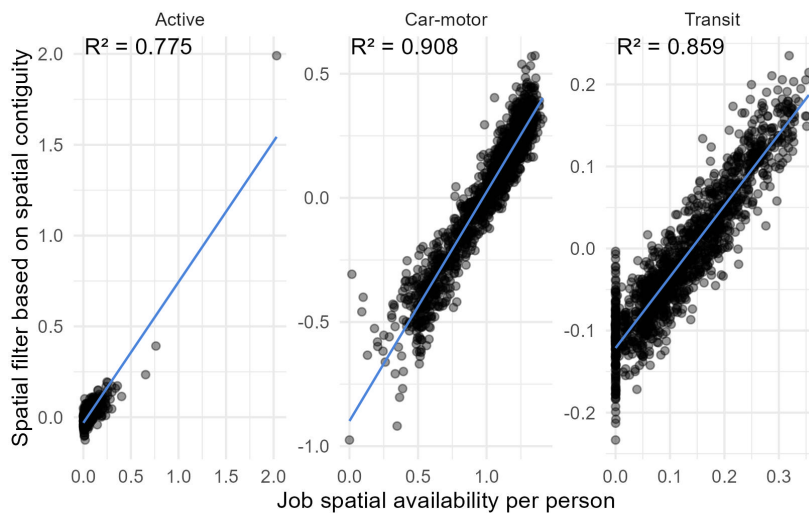
r2_labels <- spatial_filter_dist_SA_pp_3m %>%
  group_by(PwMode_label) %>%
  summarise(r2 = summary(lm(Sf_dist_SA_pp ~ SA_pp_im))$r.squared) %>%
  mutate(label = paste0("R2 = ", round(r2, 3)))

spatial_filter_dist_fig_SA_pp_3m <- ggplot(spatial_filter_dist_SA_pp_3m, aes(x = SA_pp_im, y = Sf_dist_SA_pp_3m)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "#4b84db", linewidth = 0.55) +
  geom_text(data = r2_labels,
            aes(x = -Inf, y = Inf, label = label),
            hjust = -0.1, vjust = 1.1, inherit.aes = FALSE) +
  facet_wrap(~ PwMode_label, scales = "free") +
  theme_minimal() +
  labs(x = "Job spatial availability per person", y = "Spatial filter based on spatial contiguity") +
  theme_minimal()

# Saving figure
ggsave(file = paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA_pp_3m.jpg"),
       plot = spatial_filter_dist_fig_SA_pp_3m,
       width = 14,
       height = 9,
       units = "cm",
       dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_SA_pp_3m.jpg"))

```



Visualizing the spatial filter in form of maps:

```

spatial_filter_contiguity_spatial_SA_pp_3m <- grid_expanded %>%
  filter(PwMode_label %in% (spatial_filter_dist_SA_pp_3m$PwMode_label)) %>%
  left_join(spatial_filter_dist_SA_pp_3m, by = c("CTUID" = "CTUID",

```



```

                                "PwMode_label" = "PwMode_label")) %>%
left_join(ct_file %>% mutate(CTUID = as.character(CTUID)), by = c("CTUID" = "CTUID"))

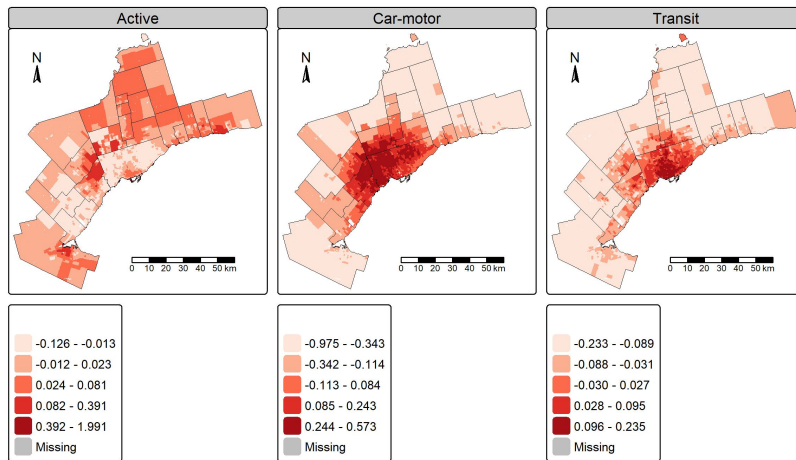
spatial_filter_contiguity_spatial_SA_pp_3m <- st_as_sf(spatial_filter_contiguity_spatial_SA_pp_3m)

spatial_filter_contiguity_fig_SA_pp_3m <- spatial_filter_contiguity_spatial_SA_pp_3m %>%
  tm_shape(spatial_filter_contiguity_spatial_SA_pp_3m) +
  tm_polygons("Sf_dist_SA_pp",
              style = "jenks",
              palette = "Reds",
              title = " ",
              border.col = NULL) +
  tm_facets(by = "PwMode_label", ncol = 3, free.scales = TRUE) +
  tm_scalebar(position = c("right", "bottom")) +
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(spatial_filter_contiguity_fig_SA_pp_3m,
          paste0(diretorio_export_figures,"contiguity_spatial_filter_SA_pp_3m.jpg"),
          width = 20,
          height = 18,
          units = "cm",
          dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"/contiguity_spatial_filter_SA_pp_3m.jpg"))

```



Hansen-type

```

modes <- unique(HT_final$PwMode_label)

mode_results_dist <- list()

for (mode in modes) {
  cat("Creating the spatial filter (distance) for the transportation mode:", mode, "\n")

  # Creating a df with the accessibility values
  y <- HT_final %>%
    filter(PwMode_label == mode) %>%
    left_join(weight_matrix_eigv_dist, by = c("CTUID" = "CTUID")) %>%
    mutate(x = 1, Sf_dist_HT = 0)

  # Creating the spatial filter (Sf_dist_HT)
  for (ev_name in names(y)[startsWith(names(y), "EV")]) {
    formula <- as.formula(paste("HT_im ~ x + Sf_dist_HT +", ev_name))
    model <- lm(formula, data = y)
    coefs <- summary(model)$coefficients

    if (ev_name %in% rownames(coefs)) {
      p_value <- coefs[ev_name, "Pr(>|t|)"]
    }
  }
}

```

```

    B_value <- coefs[ev_name, "Estimate"]

    if (!is.na(p_value) && p_value < 0.05) {
      y$Sf_dist_HT <- y$Sf_dist_HT + B_value * y[[ev_name]]
    }
  }
}

mode_results_dist[[mode]] <- y
}

## Creating the spatial filter (distance) for the transportation mode: Bike
## Creating the spatial filter (distance) for the transportation mode: Walk
## Creating the spatial filter (distance) for the transportation mode: Car-motor
## Creating the spatial filter (distance) for the transportation mode: Transit
## Creating the spatial filter (distance) for the transportation mode: Active
## Creating the spatial filter (distance) for the transportation mode: General

Creating a data frame with all spatial filters and accessibility:

spatial_filter_dist_HT <- as.data.frame(bind_rows(mode_results_dist, .id = "PwMode_label")) %>%
  dplyr::select(CTUID, PwMode_label, HT_im, Sf_dist_HT)

Pearson correlation between job spatial availability and spatial filter
by transportation mode:

correlations <- spatial_filter_dist_HT %>%
  group_by(PwMode_label) %>%
  summarise(correlation = cor(HT_im, Sf_dist_HT, use = "complete.obs"))

correlations

## # A tibble: 6 x 2
##   PwMode_label correlation
##   <chr>          <dbl>
## 1 Active        0.909
## 2 Bike         0.968
## 3 Car-motor    0.957
## 4 General      0.960
## 5 Transit      0.934
## 6 Walk         0.869

Creating a scatter plot figure:

r2_labels <- spatial_filter_dist_HT %>%
  group_by(PwMode_label) %>%
  summarise(r2 = summary(lm(Sf_dist_HT ~ HT_im))$r.squared) %>%

```

```

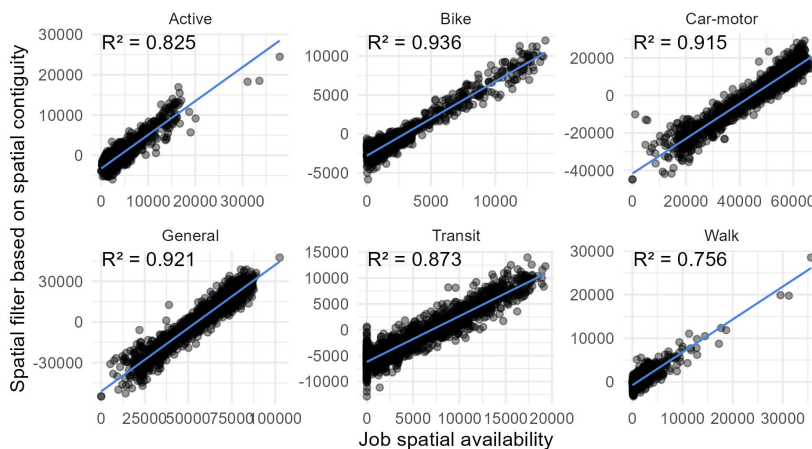
mutate(label = paste0("R² = ", round(r2, 3)))

spatial_filter_dist_fig <- ggplot(spatial_filter_dist_HT, aes(x = HT_im, y = Sf_dist_HT)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, color = "#4b84db", linewidth = 0.55) +
  geom_text(data = r2_labels,
            aes(x = -Inf, y = Inf, label = label),
            hjust = -0.1, vjust = 1.1, inherit.aes = FALSE) +
  facet_wrap(~ PwMode_label, scales = "free") +
  theme_minimal() +
  labs(x = "Job spatial availability", y = "Spatial filter based on spatial contiguity") +
  theme_minimal()

# Saving figure
ggsave(file = paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_HT.jpg"),
       plot = spatial_filter_dist_fig,
       width = 16,
       height = 9,
       units = "cm",
       dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures, "/spatial_filter_dist_scatterplot_HT.jpg"))

```



Visualizing the spatial filter in form of maps:

```

spatial_filter_contiguity_spatial_HT <- grid_expanded %>%
  filter(!is.na(PwMode_label)) %>%
  left_join(spatial_filter_dist_HT, by = c("CTUID" = "CTUID",
                                           "PwMode_label" = "PwMode_label")) %>%
  left_join(ct_file %>% mutate(CTUID = as.character(CTUID)), by = c("CTUID" = "CTUID"))

spatial_filter_contiguity_spatial_HT <- st_as_sf(spatial_filter_contiguity_spatial_HT)

```

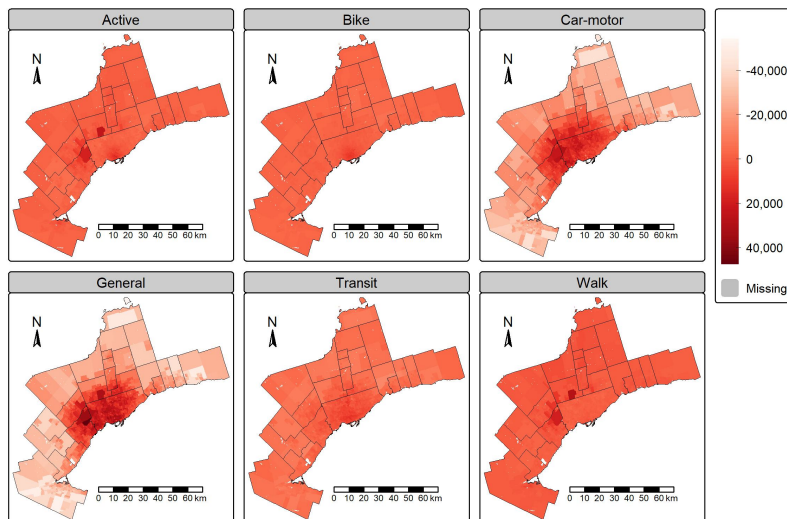
```

spatial_filter_contiguity_fig <- tm_shape(spatial_filter_contiguity_spatial_HT) +
  tm_polygons("Sf_dist_HT",
    style = "cont",
    palette = "Reds",
    title = " ",
    border.col = NULL) +
  tm_facets(by = "PwMode_label") +
  tm_scalebar(position = c("right", "bottom")) +
  tm_compass(position = c("left", "top"), size = 1.0) +
  tm_shape(census_subdivisions) +
  tm_borders("black", lwd=0.5)

tmap_save(spatial_filter_contiguity_fig,
  paste0(diretorio_export_figures,"contiguity_spatial_filter_HT.jpg"),
  width = 20,
  height = 18,
  units = "cm",
  dpi = 300)

knitr::include_graphics(paste0(diretorio_export_figures,"/contiguity_spatial_filter_HT.jpg"))

```



Data export

Creating a unique file with all spatial filters (contiguity, median commute duration, and impedance values):

```
spatial_filter <- spatial_filter_dist %>%
  left_join(spatial_filter_dist_SA_pp, by = c("CTUID", "PwMode_label")) %>%
  left_join(spatial_filter_dist_HT, by = c("CTUID", "PwMode_label"))
```

Exporting the spatial filters (original file):

```
write.csv(spatial_filter, paste0(diretorio_export_tables, "spatial_filters.csv"), row.names=FALSE)
```

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